Evaluating Gradient Agreement for Learning-Rate Control

A Comparison of Fixed, Scheduled, and Gradient-Guided Stochastic Gradient Descent

Abstract

Gradient agreement was investigated as a signal for adjusting the global learning rate of stochastic gradient descent (SGD). A fully connected neural network was trained to classify MNIST handwritten digits using fixed-rate SGD, scheduled SGD, and a controller based on cosine similarity between consecutive mini-batch gradients. Five matched random seeds were compared over 10 epochs, with shared initial weights, data splits, and batch orders within each seed. Mean final validation accuracy was 97.95% for fixed SGD, 98.12% for scheduled SGD, and 97.97% for guided SGD. Mean elapsed time to the first recorded 97% validation checkpoint was 5.73, 5.78, and 7.38 seconds, respectively. All three methods reached the target at the same epoch within each seed. Thus, the guided policy showed no observed reduction in updates to the target and required 28.7% more elapsed time than fixed SGD. The test set had been evaluated before the final five-seed comparison; the primary findings therefore represent exploratory validation evidence. Under the tested conditions, the controller did not demonstrate a practical advantage over the SGD baselines.

Keywords: stochastic gradient descent, gradient agreement, learning-rate control, neural networks, MNIST

Introduction

Learning-rate selection determines the scale of each SGD parameter update. This study examined whether agreement between consecutive mini-batch gradients could provide a useful signal for selecting that scale. The intervention compared gradient directions and adjusted one learning rate for all trainable parameters.

Research question. For a neural network trained on the MNIST handwritten digit dataset, does adjusting the stochastic gradient descent learning rate based on gradient agreement help it learn its weights faster or achieve higher accuracy?

The study was motivated by gradient-alignment approaches such as MGUP, which ranks parameters using momentum-gradient alignment and applies different update scales (Chang & Yuan, 2025). The present experiment instead used a global consecutive-gradient rule with plain SGD. Its effectiveness was assessed using validation accuracy, update counts, and elapsed time.

Method

Data, Architecture, and Training Procedure

MNIST handwritten digits (LeCun et al., 1998) were loaded through torchvision. The 60,000-image training split was partitioned into 48,000 training and 12,000 validation images using a seeded random split. Images were converted to tensors with values in [0, 1]. A separate set of 10,000 test images was used in the earlier test evaluation.

Table 1

Experimental Configuration

Component	Specification
Neural network	784 input values; 256 hidden units; ReLU; 10 output scores
Trainable parameters	203,530 weights and biases
Loss and optimizer	Cross-entropy; SGD without momentum
Training budget	Batch size 64; 10 epochs; 750 updates per epoch
Random seeds	42, 123, 456, 789, 2026
Within-seed controls	Shared split, initial weights, batch order, architecture, and budget
Validation measurement	Accuracy evaluated once at the end of each epoch
Execution environment	Mac; PyTorch 2.14.1 recorded in saved configurations

Learning-Rate Conditions

Fixed SGD used a learning rate of 0.50 throughout training. Scheduled SGD used S13: 0.50 for epochs 1-7 and 0.25 for epochs 8-10. Guided SGD compared the current gradient vector with the previous gradient vector before each update. All weights and biases were included in the flattened vectors.

Table 2

Gradient-Guided Learning-Rate Rule

Similarity condition	Selected rate
Strong agreement: s > 0.50	0.50
Weak nonnegative agreement: 0 <= s <= 0.50	0.375
Opposing directions: s < 0	0.50
First update or undefined similarity	0.50

Note. The opposing-direction rate was changed to 0.50 as an experimental revision, assigning the same rate to strong and opposing directions. Gradient agreement describes direction and does not indicate prediction correctness.

Method (continued)

Controller Implementation and Experimental Sequence

Cosine similarity was calculated as the dot product of the current and previous gradient vectors divided by the product of their norms. A detached previous-gradient vector was retained across batches and epoch boundaries. After selecting the rate, the current gradient was used in the standard SGD update. The base rate was used when no previous gradient existed or similarity was undefined.

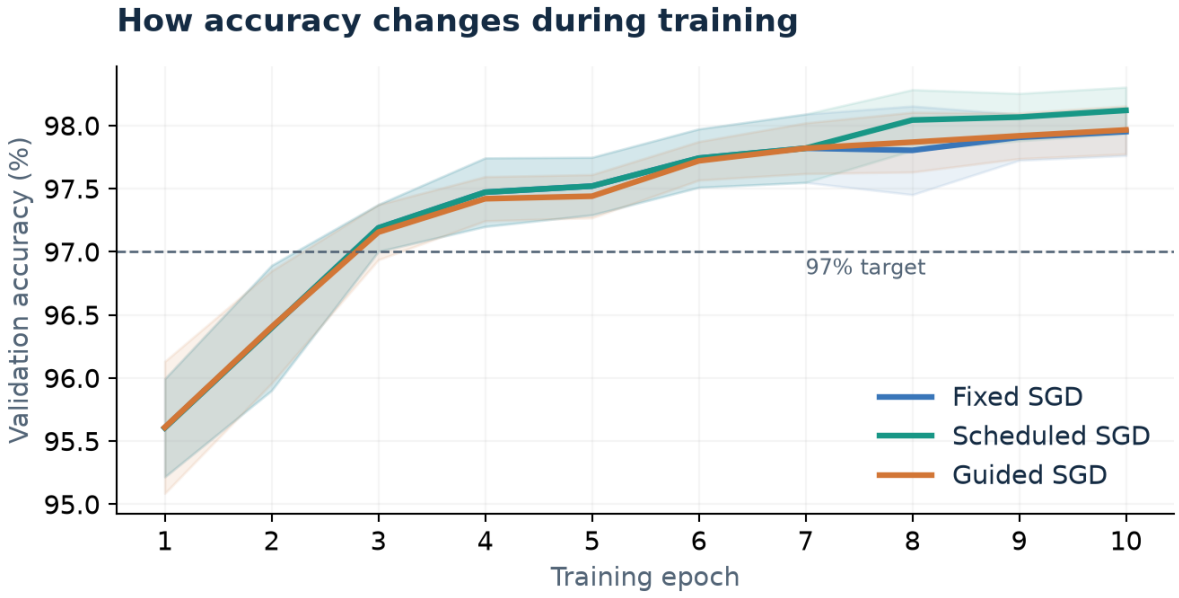
Exploration included an initial 128-unit network, fixed-rate and scheduled-rate sweeps, gradient-direction analysis, and guided-rate trials. The main comparison used the 256-unit network and five matched seeds. Test evaluation occurred before the final five-seed comparison. The analysis compares the selected configurations as an exploratory study.

Outcome Measures and Analysis

Final validation accuracy was measured after epoch 10. Time to target was defined as the cumulative elapsed time at the first recorded validation checkpoint at or above 97%. Timing included training, validation, and in-loop recording, with controller calculations included for guided SGD. Data preparation and final checkpoint serialization were excluded. Means and sample standard deviations were calculated across seeds from the saved epoch metrics; training was not rerun for this analysis.

Figure 1

Validation Accuracy Across Training Epochs



Note. Lines show means across five seeds; shaded regions show one sample standard deviation. The dashed line marks the 97% target. Validation was measured once per epoch.

Results

Final Validation Accuracy

Mean final validation accuracy was 97.95% for fixed SGD, 98.12% for scheduled SGD, and 97.97% for guided SGD (Table 3). The guided-minus-fixed difference was 0.013 percentage points. Scheduled SGD exceeded fixed SGD by 0.168 percentage points and guided SGD by 0.155 percentage points. Differences were calculated from unrounded values.

Table 3

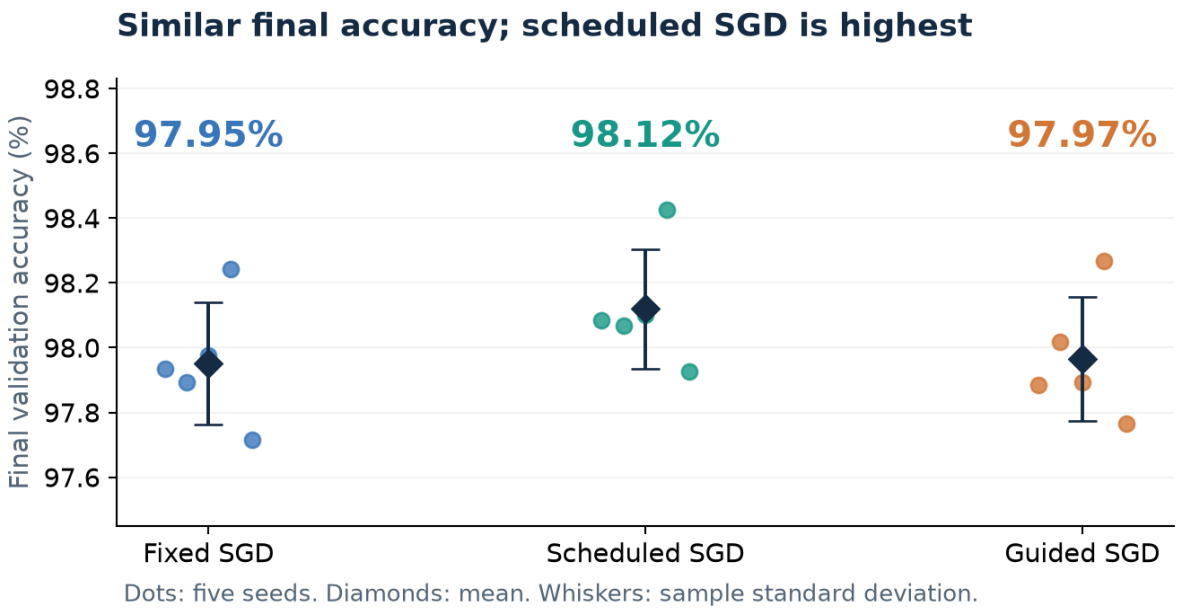
Validation Accuracy and Elapsed Time Across Five Seeds

Method	Final accuracy (%)	Time to 97% (s)	Total time (s)
Fixed SGD	97.95 (0.19)	5.73 (0.77)	17.84 (0.56)
Scheduled SGD	98.12 (0.18)	5.78 (0.77)	18.10 (0.71)
Guided SGD	97.97 (0.19)	7.38 (0.92)	22.78 (0.67)

Note. Values are means with sample standard deviations in parentheses. Accuracy variation is expressed in percentage points. Total time covers the 10-epoch training loop and validation.

Figure 2

Final Validation Accuracy by Method and Seed



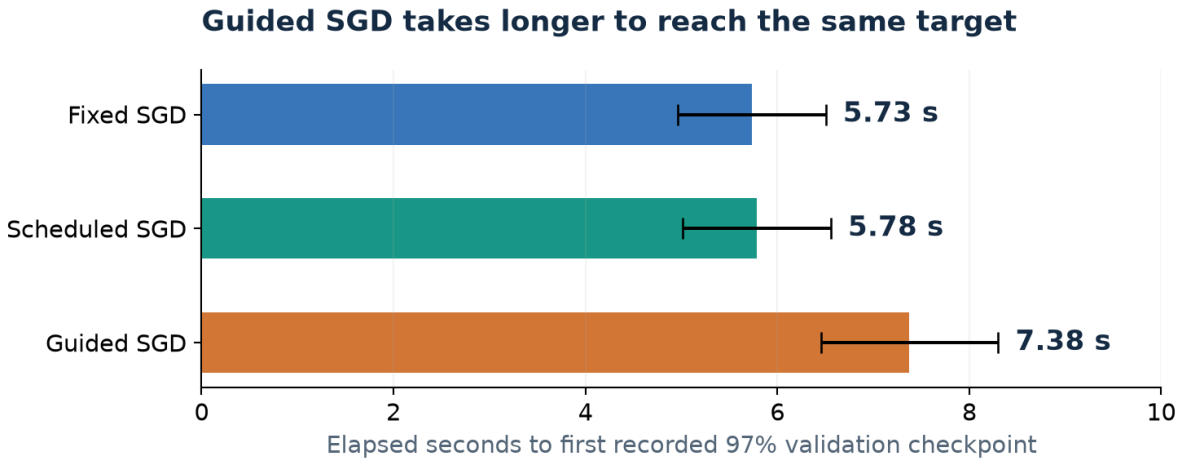
Note. Colored dots represent individual seeds. Diamonds show means and whiskers show one sample standard deviation. The vertical axis is restricted to show the small observed differences.

All runs completed 7,500 parameter updates. The guided policy produced similar observed final accuracy to fixed SGD, while scheduled SGD had the highest mean. These descriptive comparisons do not establish statistical equivalence or broad superiority.

Results (continued)

Figure 3

Elapsed Time to the First Recorded 97% Validation Checkpoint



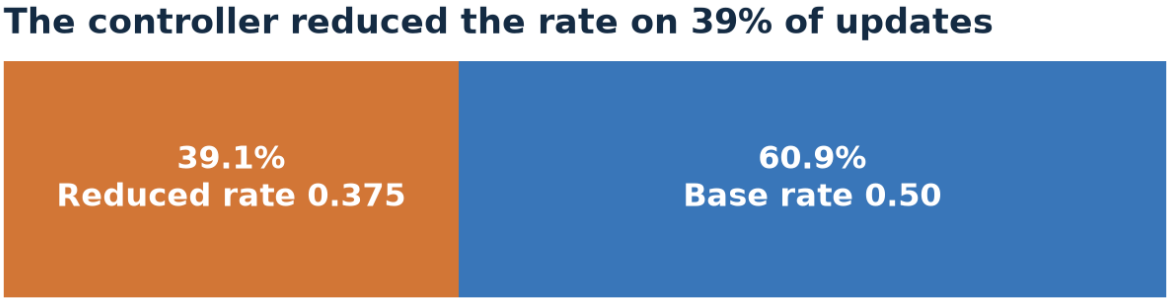
Note. Bars show means across five seeds and whiskers show one sample standard deviation. These are checkpoint times rather than exact threshold-crossing times.

Guided SGD required 7.38 seconds on average to reach the target, compared with 5.73 seconds for fixed SGD and 5.78 seconds for scheduled SGD. Relative to fixed SGD, guided SGD required 1.64 additional seconds, or 28.7% more time. Total elapsed time was 22.78 seconds for guided SGD and 17.84 seconds for fixed SGD, a 27.7% increase.

For every method, seed 123 first reached 97% at epoch 4 (3,000 updates); the remaining seeds reached the target at epoch 3 (2,250 updates). No reduction in updates to target was observed at the available measurement frequency.

Figure 4

Learning-Rate Selections Across Guided Updates



Note. Across 37,500 updates, 14,667 used 0.375 for weak nonnegative agreement, 22,828 used 0.50 for negative agreement, and five initial updates used the 0.50 fallback. No strong-agreement update was recorded.

The strong-agreement branch was unused in the five main guided runs. The observed policy therefore reduced the rate only for weak nonnegative agreement. This describes the recorded decisions; it does not establish that gradient dependence itself caused the performance difference.

Discussion

Interpretation of the Main Comparison

The tested gradient-guided policy produced nearly the same mean validation accuracy as fixed SGD and required more elapsed time. The additional work included gradient-vector construction, similarity calculation, rate selection, and detailed step recording. Because these components were not profiled separately, the runtime difference cannot be attributed solely to cosine similarity. Matching the target epoch across methods also provides no evidence of improved update efficiency at the recorded checkpoints.

Earlier Test Evaluation

Table 4

Earlier Test Results for Seed-42 Checkpoints

Checkpoint configuration	Test accuracy (%)	Cross-entropy
Fixed SGD, rate 0.60	98.30	0.06124
Scheduled SGD, S13	98.27	0.05986
Guided SGD, threshold 0.60	98.06	0.06541

Note. Each checkpoint was evaluated on 10,000 test images. The fixed rate and guided threshold differ from those in the main five-seed comparison.

These test results were obtained before the final five-seed comparison. They characterize the listed checkpoints and cannot serve as an untouched final assessment of the main configurations. Accordingly, the primary comparisons in this study use validation metrics.

Limitations

The main evidence covers one dataset, one architecture, one batch size, five seeds, and a 10-epoch budget. Seed changes affected the split, initialization, and batch order together. Several learning rates, schedules, and controller thresholds were explored during development. The reported comparison evaluates the configurations specified in the Method section.

Conclusion

Adjusting the global SGD learning rate from agreement between consecutive mini-batch gradients did not demonstrate a practical advantage in this experiment. Guided SGD achieved nearly the same final validation accuracy as fixed SGD, required more elapsed time, and reached the 97% target at the same epoch within each seed. Scheduled SGD produced the highest observed mean final validation accuracy.

A subsequent study could isolate the alignment signal by comparing the guided policy with a control that uses the same rate proportions but shuffled decisions. The same number of tuning trials could be allocated to each method, with settings selected before evaluation and tested consistently across seeds. Prior test-set use should remain documented.

Data and Reproducibility

The analysis used saved epoch metrics for 15 main runs and verified controller counts against 37,500 step records. Source-file hashes and an inventory of 155 metric and summary CSV files are retained in evidence.json. The training scripts and configurations are retained locally with the experiment outputs.

The configurable scripts use editable constants and currently specify seed 2026. Reproducing all five seeds requires changing the seed while preserving matched settings. Guided execution prevents overwriting existing runs; fixed and scheduled execution can overwrite results. A separate output copy is required to preserve the recorded experiment. Package versions are listed in requirements.txt; fresh installation and training were not verified in this analysis.

Table 5

Final Validation Accuracy and Target Epoch by Seed

Seed	Fixed (%)	Scheduled (%)	Guided (%)	Target epoch
42	97.933	98.083	97.883	3
123	97.892	98.067	98.017	4
456	97.975	98.100	97.892	3
789	98.242	98.425	98.267	3
2026	97.717	97.925	97.767	3

Note. The first recorded epoch at or above 97% was identical across all three methods within each seed.

References

Chang, D., & Yuan, G. (2025). MGUP: A momentum-gradient alignment update policy for stochastic optimization. *Advances in Neural Information Processing Systems, 38*. https://doi.org/10.52202/085713-0691

LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE, 86*(11), 2278-2324. https://bottou.org/papers/lecun-98h

Supplementary Source Map

The paths below identify the local evidence used to construct the figures and tables. In each path, {seed} denotes 42, 123, 456, 789, or 2026.

Fixed SGD metrics

output/fixed_lr_experiments/hidden_256/
seed_{seed}/LR_0.5_metrics.csv

Scheduled SGD metrics

output/scheduled_lr_experiments/hidden_256/
seed_{seed}/S13/metrics.csv

Guided SGD metrics

output/gradient_guided/hidden_256/seed_{seed}/
GG_STR0.500_WEAK0.375_NEG0.500_THR0.500/
guided_epoch_summary.csv

Guided step records and configuration

guided_step_metrics.csv and experiment_config.json in the same guided-run directories

Earlier test results

output/test_evaluation/hidden_256/seed_42/test_results.csv

Training and analysis scripts

script/main.py; script/run_fixed_lr_experiments.py;
script/run_scheduled_lr_experiments.py; script/run_gradient_guided.py;
script/analyze_gradients.py; script/evaluate_test_models.py

Earlier learning materials documented an initial controller proposal. The rule reported here follows the saved configurations of the five-seed comparison.